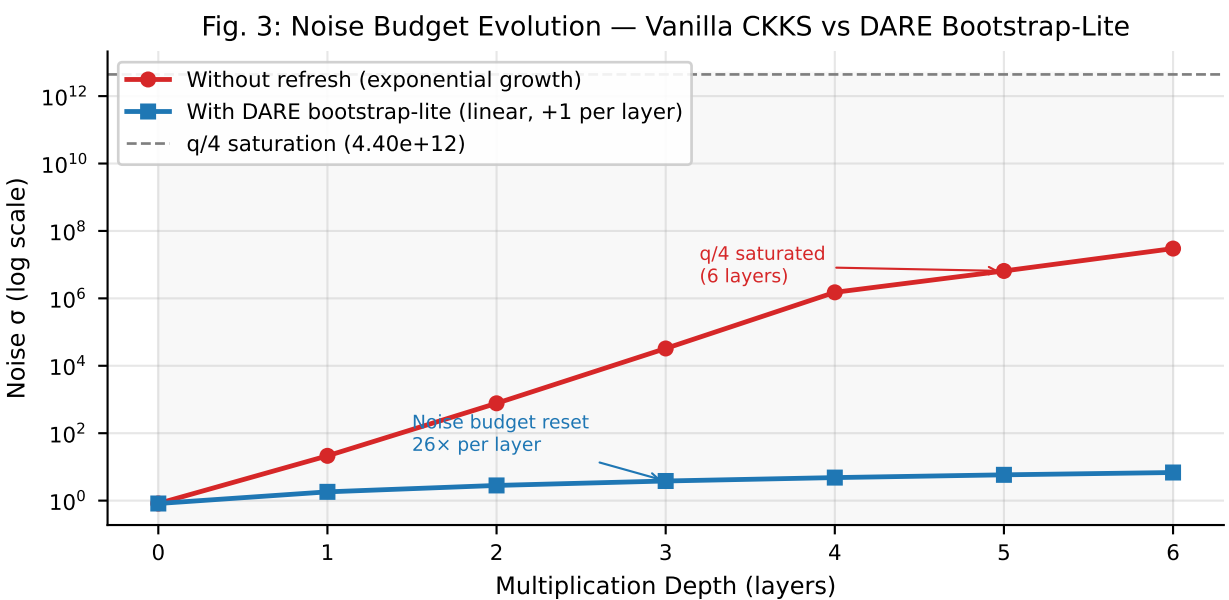
Fig. 3: Noise Budget Evolution — Vanilla CKKS vs DARE Bootstrap-Lite
Without refresh (exponential growth)
With DARE bootstrap-lite (linear, +1 per layer)
q/4 saturation (4.40e+12)
q/4 saturated
(6 layers)
Noise budget reset
26× per layer
Noise σ (log scale)
Multiplication Depth (layers)
10^{12}
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
0
1
2
3
4
5
6